AUTOMATED PARKING SYSTEM WITH LICENCE PLATE RECOGNITION AND SLOT ALLOCATION

Submitted in partial fulfillment of the requirements for the award of
Bachelor of Engineering degree in Computer Science and Engineering
with Artificial Intelligence and Machine Learning

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BONAFIDE CERTIFICATE

This is to certify that this Project Report is the bonafide work of **PUSHPARAJ BUVIGNESH (42611103)**, **PRANAY JAVVAJI (42611097),** who carried out the Project entitled "AUTOMATED PARKING SYSTEM WITH NUMBER PLATE RECOGNITION AND SLOT ALLOCATION" under my supervision from November 2025 to April 2026.

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ABSTRACT

Automated parking has become an essential requirement in modern urban environments, where space constraints and increasing vehicle numbers make manual parking inefficient and error-prone. Traditional parking systems require human supervision and often lead to mismanagement, time wastage, and an increased risk of collisions during tight parking maneuvers. Despite technological progress, integrating multiple intelligent functions such as vehicle detection, license plate recognition, collision prevention, and slot allocation into a single automated system remains a challenge. This study proposes an intelligent parking management system that leverages YOLO-based object detection for classifying vehicles as cars or bikes and EasyOCR for accurate license plate recognition. The system dynamically allocates available parking slots and maintains a real-time database storing vehicle details, slot numbers, and entry timestamps. At exit, it retrieves the record, calculates parking duration, and computes the corresponding fare. Furthermore, vehicle re-identification ensures continuous tracking throughout the parking process, while a camera calibration-based collision detection module estimates distances to walls and pillars, reducing accidents during parking. Through the integration of these technologies, the proposed system aims to enhance efficiency, safety, and accuracy in parking operations, minimize human intervention, and provide a scalable solution for smart cities, commercial complexes, and public parking facilities, ultimately supporting the vision of intelligent transportation systems.

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LIST OF ABBREVATIONS

ABBREVATION	EXPANSION
ANPR	Automatic Number Plate Recognition
API	Application Programming Interface
CNN	Convolutional Neural Network
CRR	Character Recognition Rate
CUDA	Compute Unified Device Architecture
DFL	Distribution Focal Loss
FAR	False Acceptance Rate
FPS	Frames Per Second
GPU	Graphics Processing Unit
IoU	Intersection over Union
JSON	JavaScript Object Notation
LPR	License Plate Recognition
OCR	Optical Character Recognition
Re-ID	Vehicle Re-Identification
RFID	Radio Frequency Identification
RTSP	Real-Time Streaming Protocol
UI	User Interface
V2I	Vehicle-to-Infrastructure
YOLO	You Only Look Once

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